Artificial Intelligence

# UNIVERSITY OF THE PUNJAB

**(Faculty of Computing and Information Technology)**

**Machine Learning**

# Human Activity Recognition

**(Using CogAge Dataset)**

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Submitted To

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# Assignment Report

## Sensor-based Human Activity Classification using Early Fusion on the CogAge Dataset

Student Information

**Course:** Pattern Recognition
**Assignment:** Assignment #2 – Sensor-based Human Activity Classification
**Approach Used:** Early Fusion
**Selected Subjects:** Adeel and Wang

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## 1. Abstract

Human Activity Recognition (HAR) is the task of automatically identifying human physical activities from wearable or mobile sensor data. In this assignment, an early-fusion based activity recognition pipeline was implemented using multimodal wearable sensor data from the CogAge 12 Activities dataset. The experiment was conducted on two selected subjects, Adeel and Wang. Signals from smartphone, smartwatch, and smart glasses sensors were combined using feature-level fusion. Each sensor stream was resampled to a fixed sequence length of 100 time steps and concatenated into a unified multivariate sequence of shape (100, 27). Three deep learning models were trained and evaluated: LSTM, Simple RNN, and CNN-LSTM. Experiments were performed under both subject-dependent and subject-independent settings. In the subject-dependent setting, the LSTM achieved the best performance with 81.82% test accuracy. In the subject-independent setting, the Simple RNN achieved the best performance with 22.90% test accuracy. The results show that subject-dependent recognition is significantly easier than subject-independent recognition, and cross-subject generalization remains a challenging problem.

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## 2. Introduction

Human Activity Recognition is an important application area of machine learning and pattern recognition. It is widely used in health monitoring, smart homes, assisted living, fitness tracking, and context-aware computing. Modern HAR systems rely on wearable and mobile sensors such as accelerometers and gyroscopes to capture body motion and behavior.

The CogAge dataset provides multimodal wearable sensor data collected from different devices, including smartphone, smartwatch, and smart glasses. Since these devices capture complementary motion information, combining them can improve activity recognition performance. One common approach for combining multimodal data is **Early Fusion**.

**Definition:**
**Early Fusion** is a multimodal fusion technique in which signals or features from multiple sensors are combined at the input level before being passed to the classifier.

In this assignment, early fusion was applied to multimodal sensor data, and three sequential deep learning models were compared in both subject-dependent and subject-independent experimental settings.

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## 3. Dataset Description

The dataset used in this assignment was the **CogAge Sensor-based Human Activities Dataset**. The experiments were carried out on the following 12 activities:

1. Bending
2. Bring
3. CleanFloor
4. CleanSurface
5. Drink
6. EatSmallThing
7. DryOffHands
8. Standing
9. Sitting

10. Walking
11. Lying
12. Squatting

For this work, only **two subjects** were selected:

- Adeel
- Wang

Each activity folder contained multiple event files recorded from different devices and sensors.

Sensors used

The following 9 sensor streams were used for early fusion:

**Smart glasses (Jins):**

- JinsAccelerometer
- JinsGyroscope

**Smartphone:**

- PhoneAccelerometer
- PhoneGyroscope
- PhoneGravity
- PhoneLinearAccelerometer
- PhoneMagnetometer

**Smartwatch:**

- android.sensor.accelerometer
- android.sensor.gyroscope

Since each sensor contains three axes (x, y, z), the total number of features after fusion became:

**9 sensors × 3 axes = 27 features per time step**

## 4. Methodology

### 4.1 Data Preprocessing

The raw sensor files had different sequence lengths. For example:

- glasses signals were approximately 100 samples long
- phone signals were approximately 1000 samples long
- watch signals were approximately 494 samples long

Because early fusion requires aligned sequences, all sensor streams were converted to the same temporal length.

**Definition:**
**Resampling** is the process of converting a variable-length sequence into a fixed-length sequence while preserving its temporal pattern as much as possible.

In this assignment, each sensor stream was resampled to **100 time steps** using interpolation. After resampling, the 9 sensor streams of the same event were concatenated along the feature axis.

Thus, one fused sample had the shape:

**(100, 27)**

where:

- 100 = time steps
- 27 = fused features

### 4.2 Label Encoding

The activity names were converted from string labels into numeric class indices so that they could be used for neural network training.

## 4.3 Experimental Settings

Two evaluation settings were used:

*Subject-Dependent Setting*

In this setting, data from both selected subjects were included in the dataset, and the samples were split into training, validation, and testing sets.

**Split used:**

- 70% training
- 15% validation
- 15% testing

*Subject-Independent Setting*

In this setting, the model was trained on one subject and tested on another unseen subject.

**Configuration used:**

- Training subject: **Adeel**
- Testing subject: **Wang**

A validation split was created only from the training subject data.

## 4.4 Models Used

Three deep learning models were implemented and compared.

*1. LSTM*

**Definition:**
**Long Short-Term Memory (LSTM)** is a recurrent neural network architecture designed to capture long-range temporal dependencies in sequential data.

*2. Simple RNN*

**Definition:**
A **Simple RNN** is a recurrent neural network that models sequence dependencies using recurrent hidden states, but it is generally less powerful than LSTM for long-term temporal information.

*3. CNN-LSTM*

**Definition:**
A **CNN-LSTM** is a hybrid model in which convolutional layers first extract local temporal features, and LSTM layers then model long-term sequential dependencies.

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## 5. Implementation Details

The models were implemented in **PyTorch** and trained on **Google Colab using Tesla T4 GPU**. Best model weights were saved to Google Drive after each epoch whenever validation accuracy improved.

Common training settings

- Optimizer: Adam
- Learning rate: 0.001
- Loss function: CrossEntropyLoss
- Number of epochs: 30
- Batch size: 32

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## 6. Results

### 6.1 Subject-Dependent Results

| Model | Best Validation Accuracy | Test Accuracy | Test Loss |
|---|---|---|---|
| LSTM | 78.46% | **81.82%** | 0.7342 |
| Simple RNN | 70.77% | 78.79% | 0.8311 |
| CNN+LSTM | 69.23% | 69.70% | 1.0338 |

Observation

The LSTM achieved the best overall performance in the subject-dependent setting. This indicates that LSTM was better able to capture the temporal dependencies of the fused multimodal sensor signals.

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### 6.2 Subject-Independent Results

**Train on Adeel, Test on Wang**

| Model | Best Validation Accuracy | Test Accuracy | Test Loss |
|---|---|---|---|
| LSTM | 75.56% | 13.55% | 3.2914 |
| Simple RNN | 60.00% | **22.90%** | 2.8820 |
| CNN+LSTM | 75.56% | 22.43% | 2.5165 |

Observation

In the subject-independent setting, the best test accuracy was achieved by the Simple RNN. However, all models performed much worse compared to the subject-dependent setting.

## 7. Discussion

The experimental results clearly show a major difference between subject-dependent and subject-independent recognition.

In the **subject-dependent setting**, the models achieved relatively high performance, especially the LSTM model with 81.82% test accuracy. This suggests that when the training and testing samples come from the same subject pool, the model can effectively learn activity-related temporal patterns.

In contrast, the **subject-independent setting** was much more challenging. Even though the LSTM and CNN-LSTM achieved relatively high validation accuracies on the training-subject validation split, their performance dropped significantly on the unseen test subject. This indicates that the models learned subject-specific motion characteristics rather than robust subject-invariant activity patterns.

Possible reasons for poor subject-independent performance include:

- different movement styles across subjects
- variation in signal magnitude and motion intensity
- device placement differences
- limited number of subjects used in the experiment
- strong inter-subject variability in wearable sensor data

These results show that cross-subject generalization remains a difficult problem in multimodal HAR.

## 8. Conclusion

This assignment implemented an early-fusion based multimodal human activity recognition pipeline using sensor data from smartphone, smartwatch, and smart glasses. Data from two selected subjects, Adeel and Wang, were used. All nine sensor streams were resampled to a fixed length of 100 time steps and fused into a multivariate sequence of shape (100, 27).

Three deep learning models, LSTM, Simple RNN, and CNN-LSTM, were evaluated in both subject-dependent and subject-independent settings.

The results showed that:

- In the **subject-dependent setting**, **LSTM** was the best model with **81.82% test accuracy**.
- In the **subject-independent setting**, **Simple RNN** was the best model with **22.90% test accuracy**.

The large performance gap between the two settings indicates that subject-independent human activity recognition is significantly more difficult than subject-dependent recognition. This is mainly due to poor cross-subject generalization and subject-specific variations in multimodal sensor data.

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## 9. Future Work

Future improvements for this work may include:

- using more subjects for training
- applying normalization per subject or per sensor stream
- using data augmentation techniques
- trying attention-based sequence models
- using leave-one-subject-out cross-validation
- exploring better domain generalization methods

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## 10. References


1. CogAge Sensor-based Human Activity Dataset
2. PyTorch Documentation
3. Course material and assignment instructions provided by the instructor